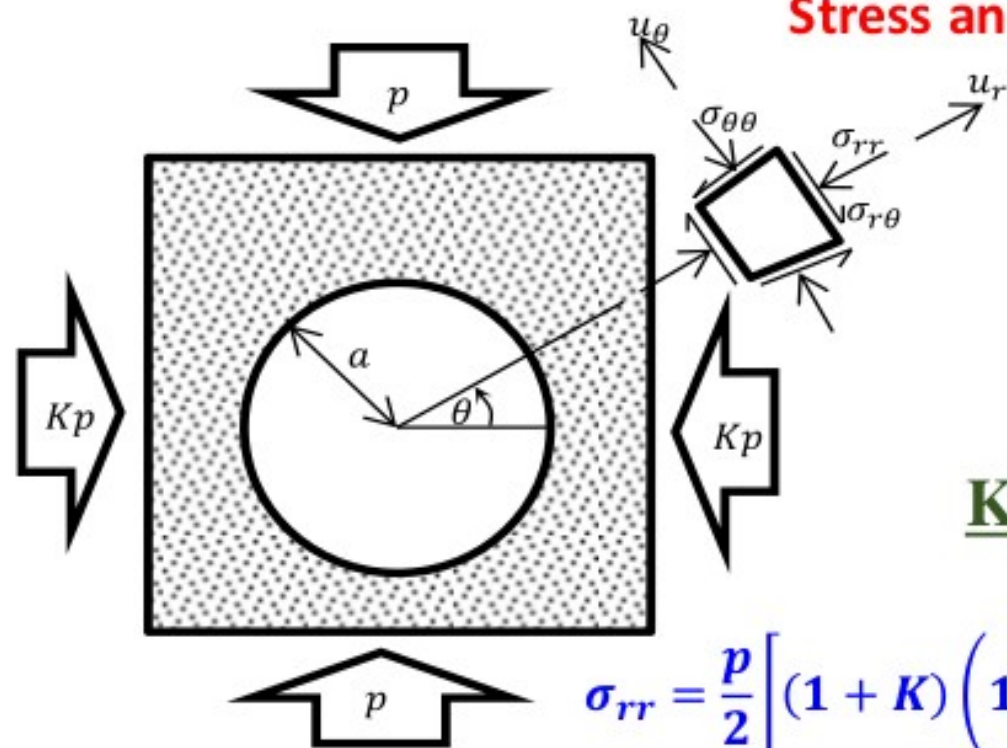


Stress and displacement distribution around a circular opening

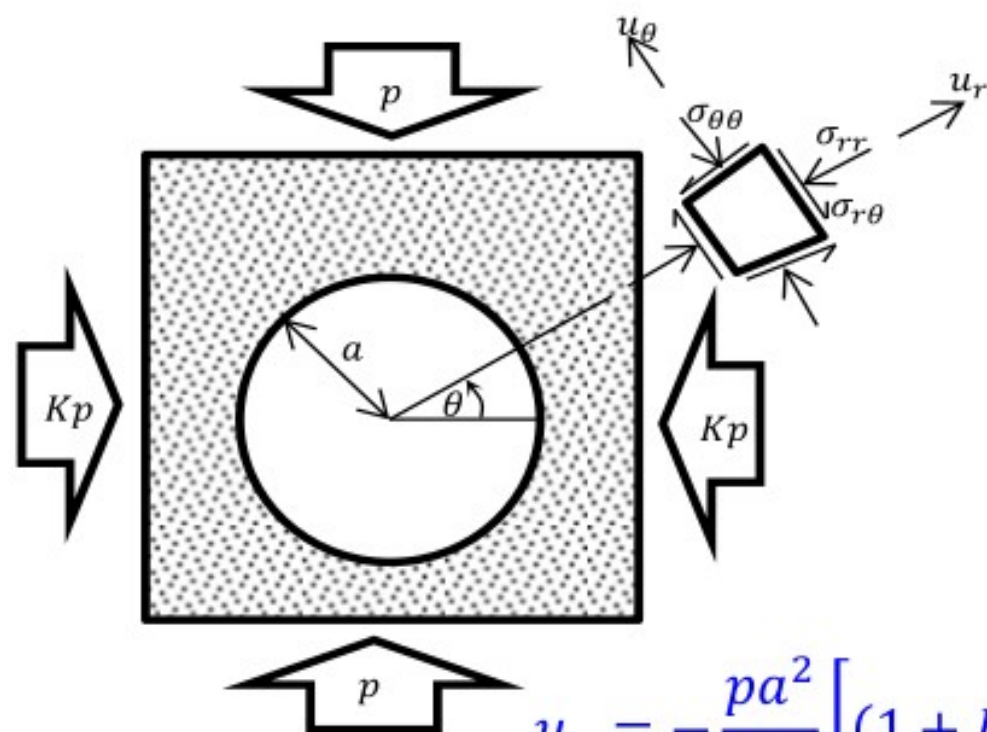


Kirsch (1898) Equations

$$\sigma_{rr} = \frac{p}{2} \left[(1 + K) \left(1 - \frac{a^2}{r^2} \right) - (1 - K) \left(1 - 4 \frac{a^2}{r^2} + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{\theta\theta} = \frac{p}{2} \left[(1 + K) \left(1 + \frac{a^2}{r^2} \right) + (1 - K) \left(1 + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right]$$

$$\sigma_{r\theta} = \frac{p}{2} \left[(1 - K) \left(1 + 2 \frac{a^2}{r^2} - 3 \frac{a^4}{r^4} \right) \sin 2\theta \right]$$

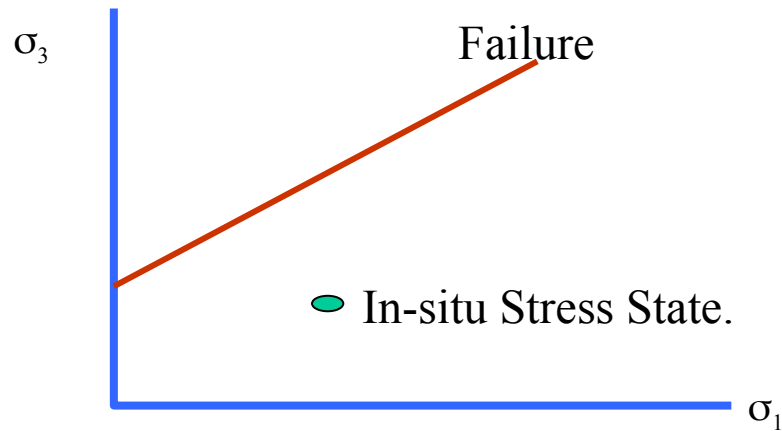


$$u_r = -\frac{pa^2}{4Gr} \left[(1+K) - (1-K) \left\{ 4(1-\nu) - \frac{a^2}{r^2} \right\} \cos 2\theta \right]$$

$$u_\theta = -\frac{pa^2}{4Gr} \left[(1-K) \left\{ 2(1-2\nu) + \frac{a^2}{r^2} \right\} \sin 2\theta \right]$$

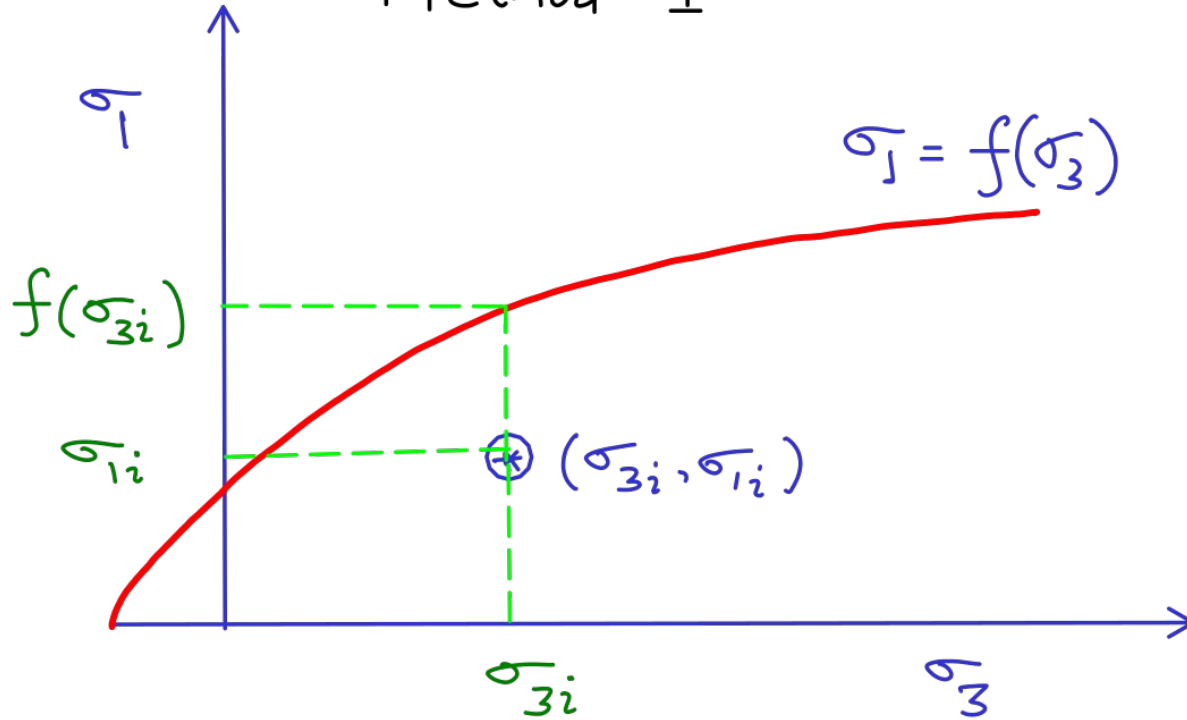
Why measure in-situ Stresses ?

- Engineering analyses require boundary conditions
- One of the most **important** boundary conditions for the analysis of underground excavations is the **in-situ stress**.



$$\text{FOS} = \frac{\text{Strength}}{\text{Stress}}$$

Method-I

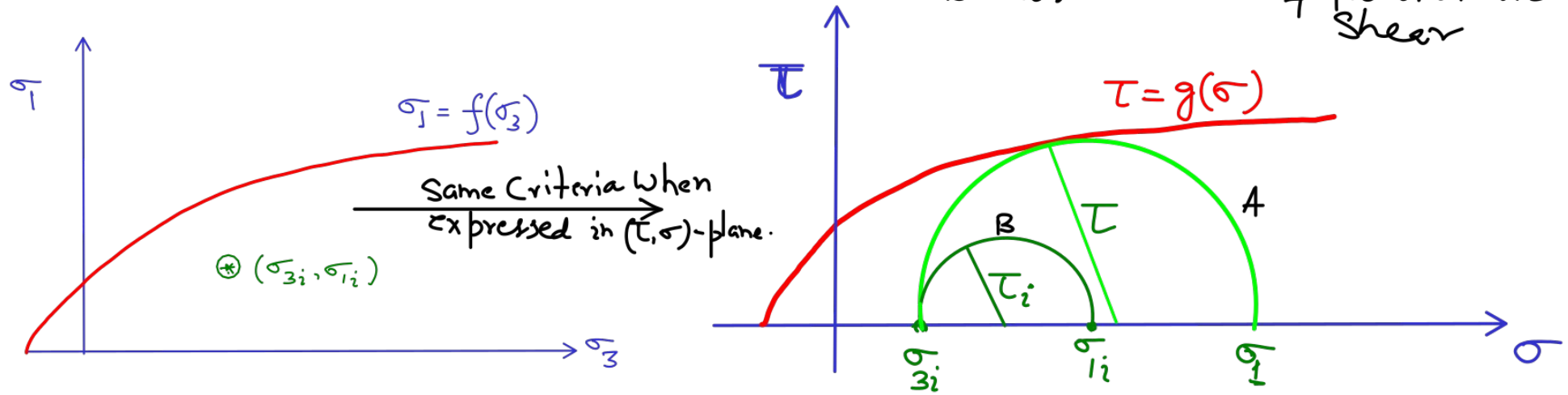


$$\begin{aligned} FOS &= \frac{\text{Strength}}{\text{Stress}} \\ &= \frac{f(\sigma_{3i})}{\sigma_{1i}} \end{aligned}$$

However, when $-\sigma_{3i} > \sigma_t$
The

$$FOS = -\frac{\sigma_t}{\sigma_{3i}}$$

Method-II (Mechanically More Sound) $\rightarrow$ Because the mode of failure is in shear



$$FOS = \frac{\text{Shear Strength}}{\text{Shear Stress}} = \frac{\text{Radius of Circle A}}{\text{Radius of Circle B}} = \frac{\sigma_1 - \sigma_{3i}}{\sigma_{li} - \sigma_{3i}} = \frac{f(\sigma_{3i}) - \sigma_{3i}}{\sigma_{li} - \sigma_{3i}}$$

Except when, $-\sigma_{3i} > \sigma_t$ Then

$$FOS = -\frac{\sigma_t}{\sigma_{3i}}$$

Activities requiring Knowledge of In-situ Stresses :

Civil and Mining Engineering :

Stability of underground excavations (tunnels, mines , caverns, shafts, stopes, haulages)

Drilling and Blasting

Pillar Design

Design of Support Systems

Prediction of rock bursts.

Fluid flow and contaminant transport.

Dams

Slope Stability

Energy Development

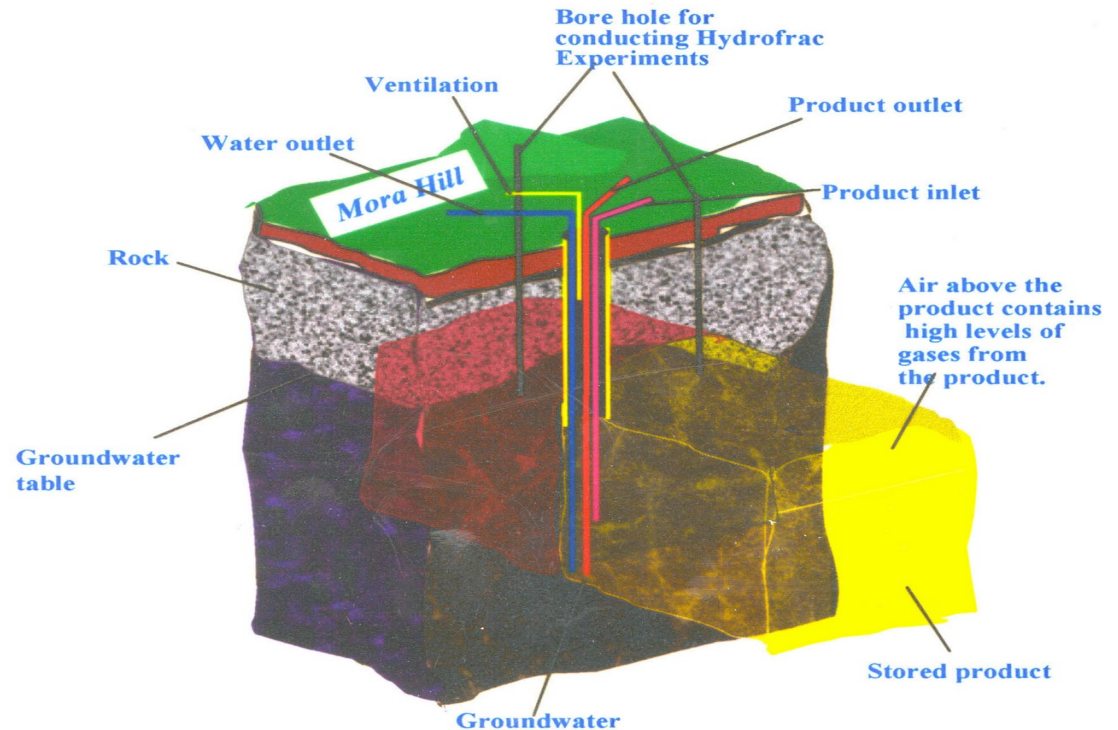
Borehole stability and deviation

Borehole deformation and failure

Fracturing and fracture Propagation

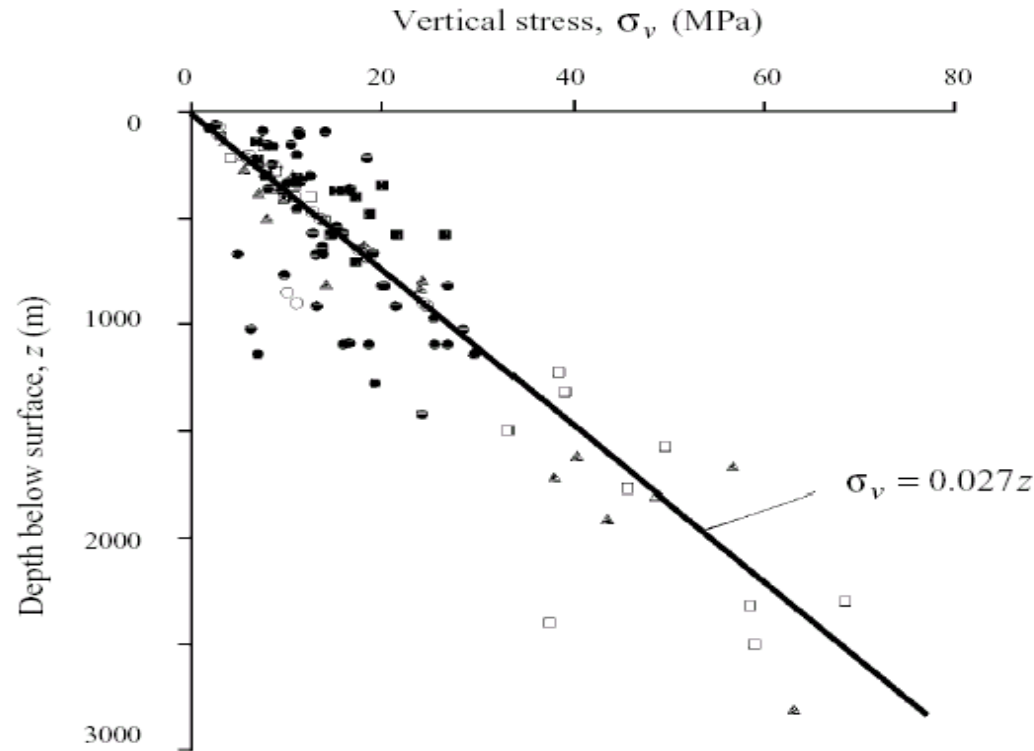
Fluid flow and geothermal problems

Reservoir production management



Activity	Stress Analysis	Deformation Analysis	Support Design
Civil Engineering			
Cavern	✓	✓	✓
Tunnel	✓	✓	✓
Pressure Shafts	✓	✗	✓
Mining Engineering			
Blasthole	✓	✗	✗
Haulage	✓	✗	✓
Pillar	✓	✗	✓
Rock Burst	✓	✗	✓
Shaft	✓	✗	✓
Stope	✓	✓	✓

Activity	Stress Analysis	Deformation Analysis	Support Design
Energy Development			
Boreholes	✓	✓	✓
Underground Excavations	✓	✓	✓
Geology / Geophysics			
Earthquake Engg.	✓	✓	✗
Plate Tectonics	✓	✓	✗
Structural Geology	✓	✓	✗
Volcanology	✓	✓	✗



Consider an element of rock at a depth of 1,000 m below the surface.

The vertical stress on the element is 2,700 tonnes/m² or 27 MPa. This

stress is estimated from the simple relationship:

$$\sigma_v = \gamma z$$

where σ_v is the vertical stress

γ is the unit weight of the overlying rock and

z is the depth below surface.

Measurements of vertical stress at various mining and civil engineering sites around the world confirm that this relationship is valid although, as illustrated in the figure there is a significant amount of scatter in the measurements.

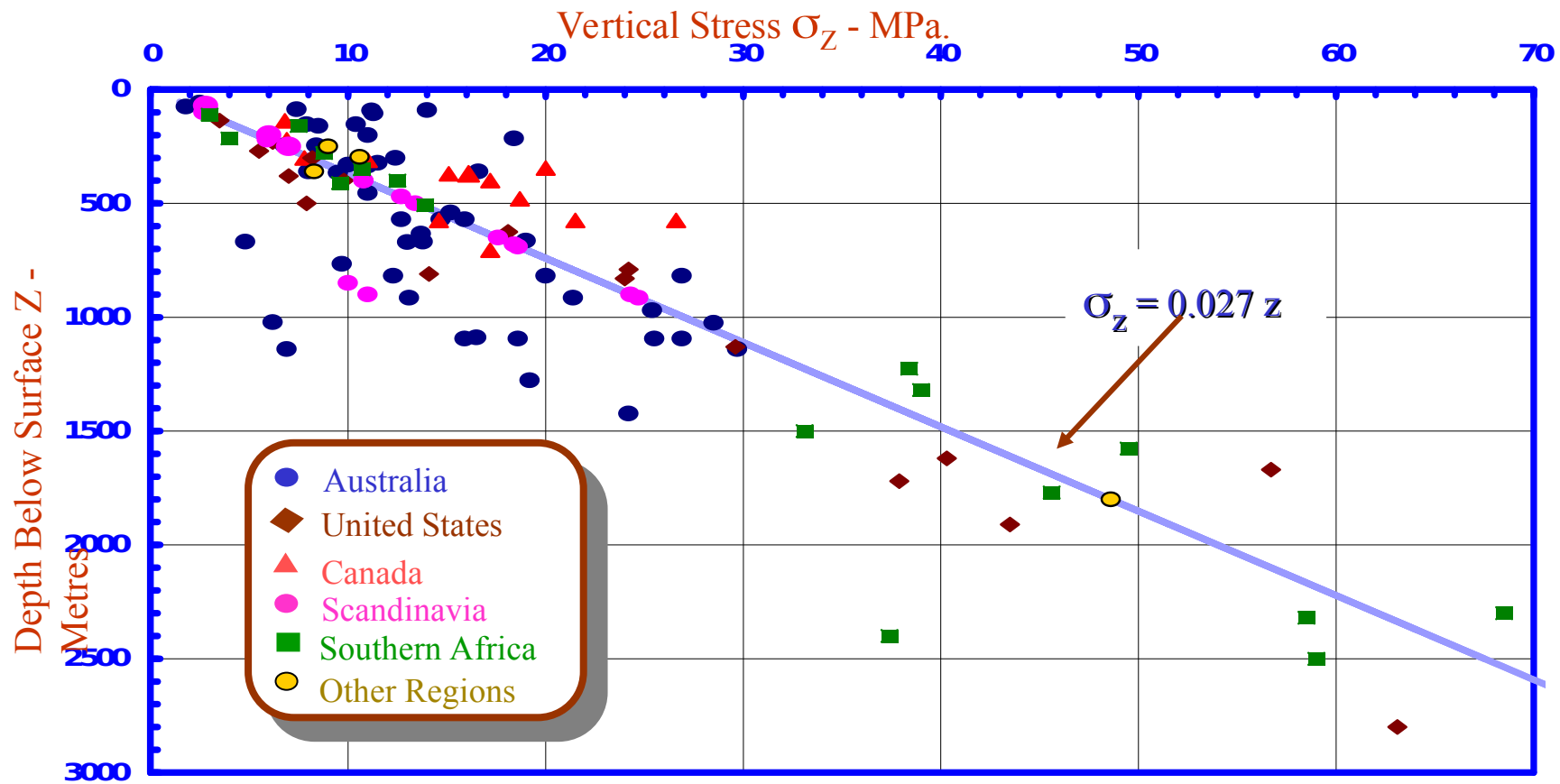
The horizontal stresses acting on an element of rock at a depth z below the surface are much more difficult to estimate than the vertical stresses. Normally, the ratio of the average horizontal stress to the vertical stress is denoted by the letter k such that

$$\sigma_h = k\sigma_v = k \gamma z$$

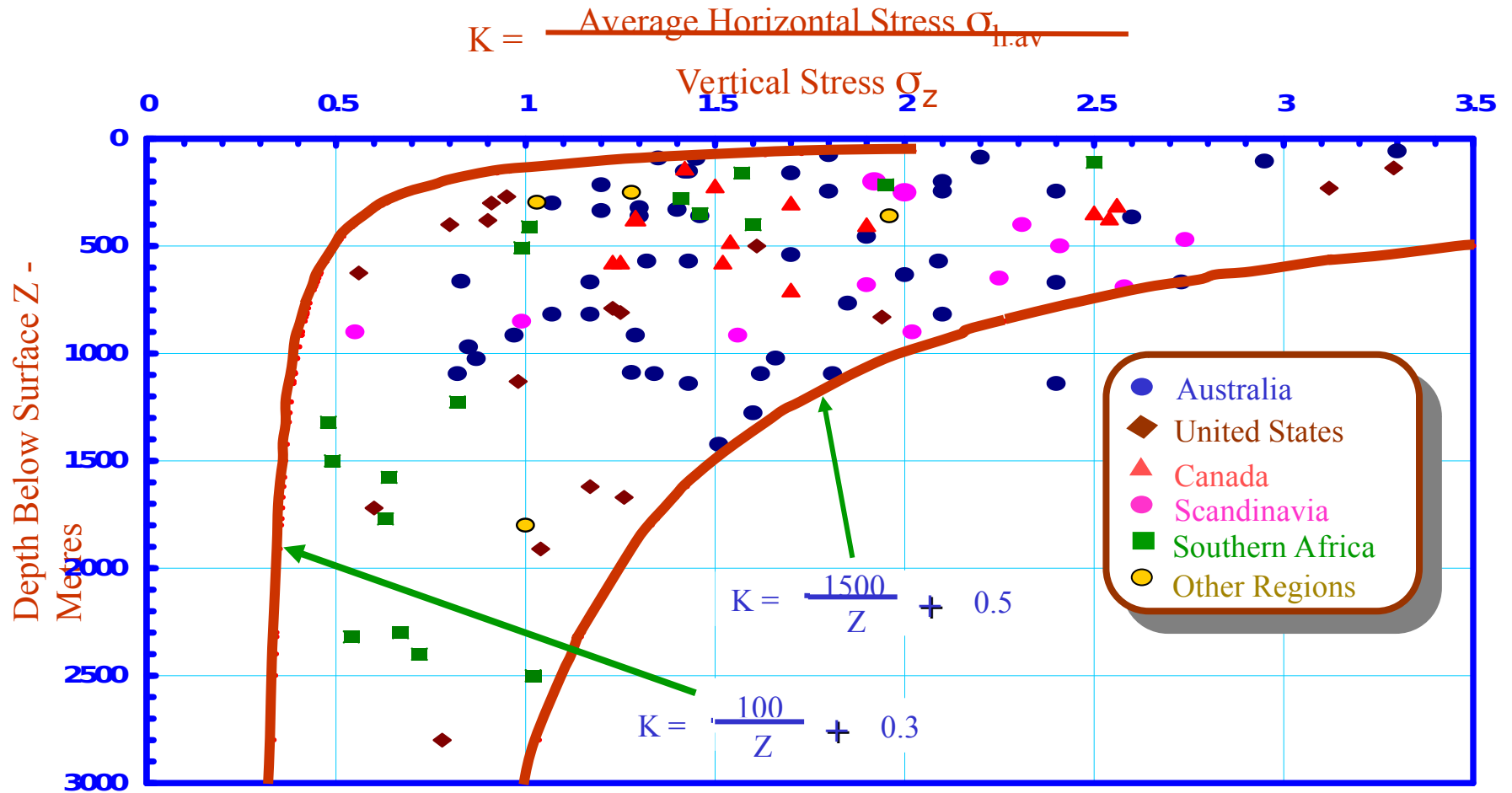
Terzaghi and Richart (1952) suggested that, the value of k is independent of depth and is given by

$$k = \nu(1 - \nu),$$

where ν is the Poisson's ratio of the rock mass.



Plot of vertical stresses against depth below surface



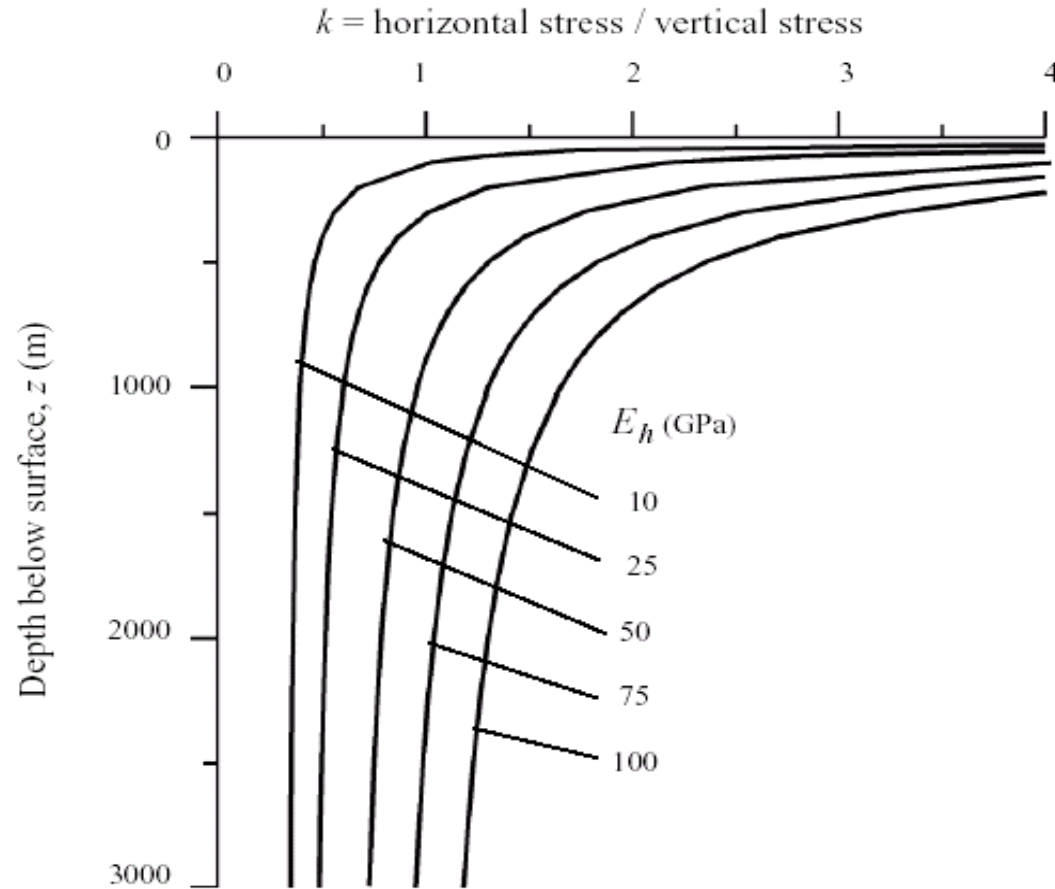
Variation of ratio of average horizontal stress to vertical stress with depth below surface.

Sheorey's (1994) Model :

- He developed an **elasto-static thermal stress model** of the earth.
- The model considers **curvature of the crust** and **variation** of elastic constants, density and thermal expansion coefficients **through the crust** and **mantle**.

$$k = 0.25 + 7 E_h \left(0.001 + \frac{1}{z} \right)$$

- Where **E_h = (GPa)** is the average deformation modulus of the upper part of the earth's crust measured in a horizontal direction.
- This direction of measurement is important particularly in layered sedimentary rocks, in which the deformation modulus may be significantly different in different directions.

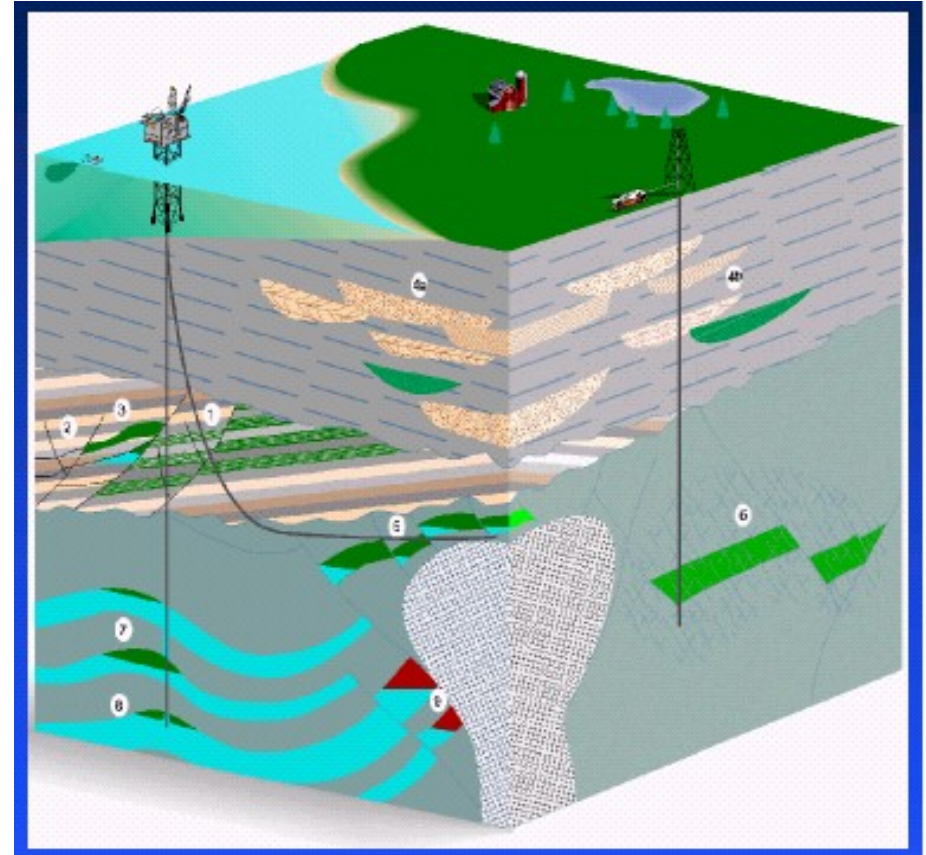


The curves relating k with depth below surface z are **similar to** those published by Brown and Hoek (1978), Herget (1988) and others for measured in situ stresses.

Hence above equation is considered to provide a **reasonable basis for estimating the value of k** .

Limitations of Sheorey's Model :

- This model does not explain the occurrence of measured vertical stresses that are **higher than the calculated overburden pressure**,
- Unable to explain the presence of very high horizontal stresses at some locations or **why the two horizontal stresses equal**. These differences are probably due to local topographic and features that cannot be taken into account in a large scale model such as by Sheorey.

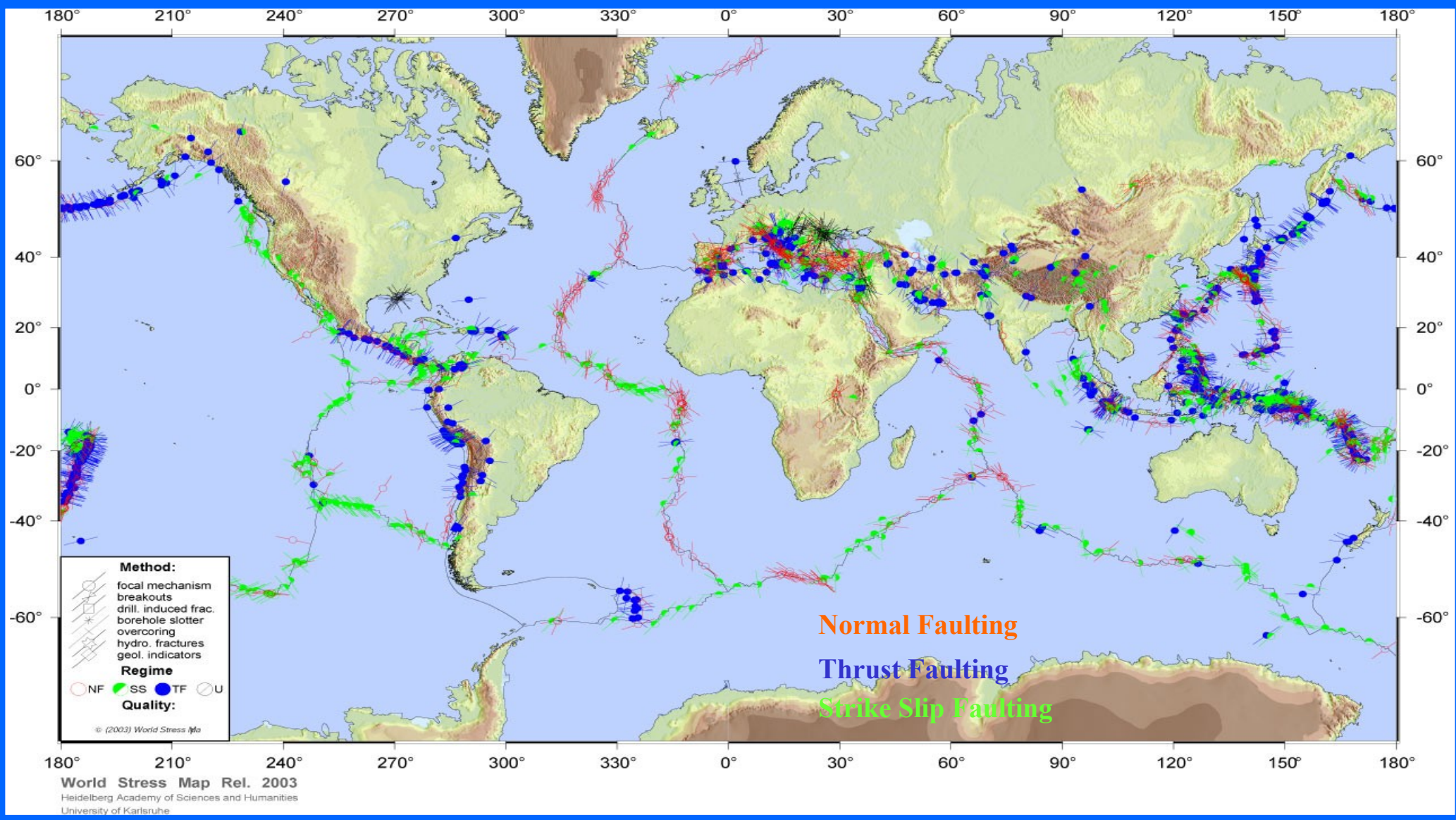


The World Stress Map project :

- Completed in July 1992, involved over 30 scientists from 18 countries
- Carried out under the auspices of the International Lithosphere Project (Zoback, 1992).

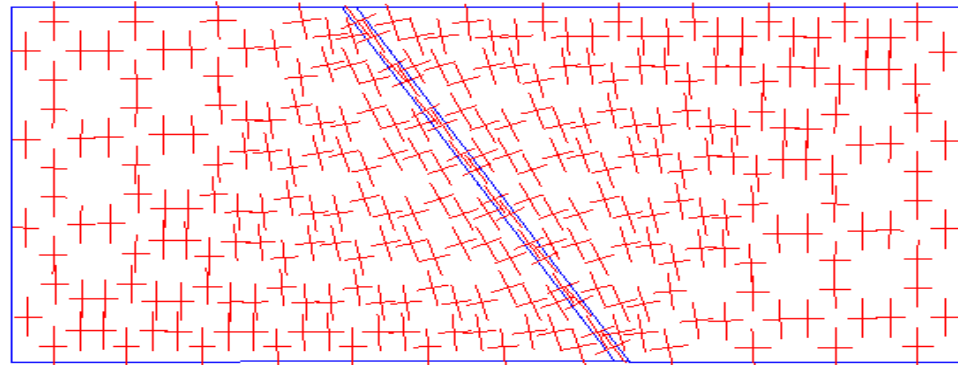
The aim of the project & Salient Features of the Data.

- The Aim was to compile a global database of contemporary tectonic stress data.
- Currently over 7,300 stress orientation entries are included in a digital database. Of these approximately 4,400 observations are considered reliable tectonic stress indicators, recording horizontal stress orientations to within $< \pm 25^\circ$.
- The data included in the World Stress Map are derived mainly from
 - Geological observations on earthquake focal mechanisms,
 - volcanic alignments and fault slip interpretations.
 - Less than 5% of the data is based upon hydraulic fracturing or overcoring measurements of the type commonly used in mining and civil engineering projects.
 - Detailed hydraulic fracturing testing in a number of boreholes beginning very close to surface (10-20 m depth) has revealed marked changes in stress orientations and relative magnitudes with depth in the upper few hundred meters, possibly related to effects of nearby topography or a high degree of near surface fracturing.



Effect of Geological Structures and Heterogeneities.

- In-situ stresses may vary substantially from one layer to another in sedimentary formations due to abrupt changes in stiffness.
- Stress jumps and non-homogeneous stress fields can also occur while crossing discontinuities, dikes, faults and folds.
- this example shows the stress distribution in a 6000m X 4000m rock mass consisting of two different blocks. Despite the uniform boundary conditions applied at the boundary of the domain, non uniform stresses develop in each block of the rock mass, as sliding takes place along the block boundaries.



JOB TITLE :

UDEC (Version 3.00)

LEGEND

4-Jul-03 11:45

cycle 5040

time 4.712E+01 sec

block plot

XX stress contours

contour interval= 2.000E+06

-2.400E+07 to -1.200E+07

-2.400E+07

-2.200E+07

-2.000E+07

-1.800E+07

-1.600E+07

-1.400E+07

-1.200E+07

principal stresses

minimum = -8.226E+07

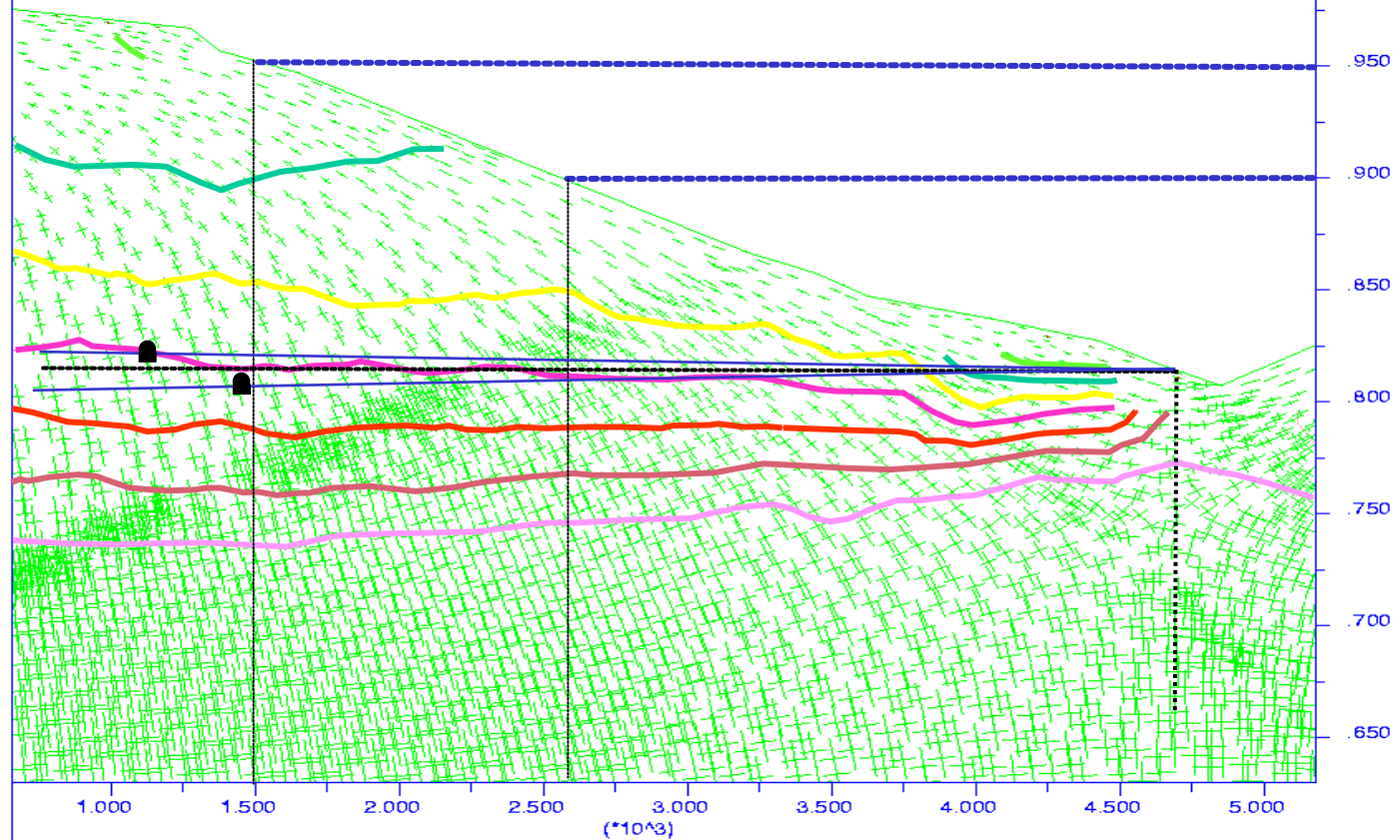
maximum = 1.912E+06

0

5E 8

nirm
kgf

Effect of Topography



JOB TITLE : Vertical Stress

UDEC (Version 3.00)

LEGEND

4-Jul-03 11:45

cycle 5040

time 4.712E+01 sec

block plot

YY stress contours

contour interval= 2.000E+07

-1.600E+08 to -2.000E+07

-1.600E+08

-1.400E+08

-1.200E+08

-1.000E+08

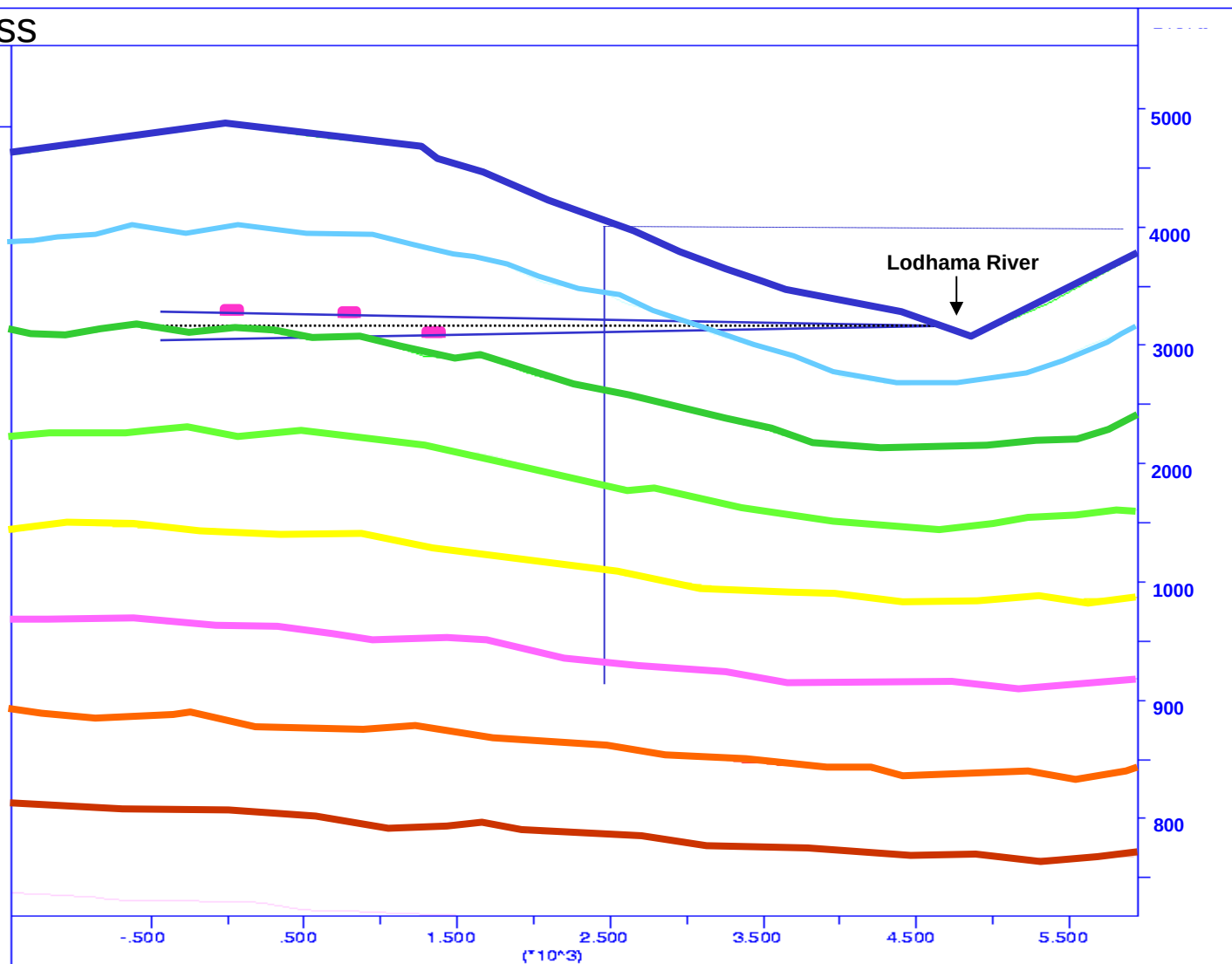
-8.000E+07

-6.000E+07

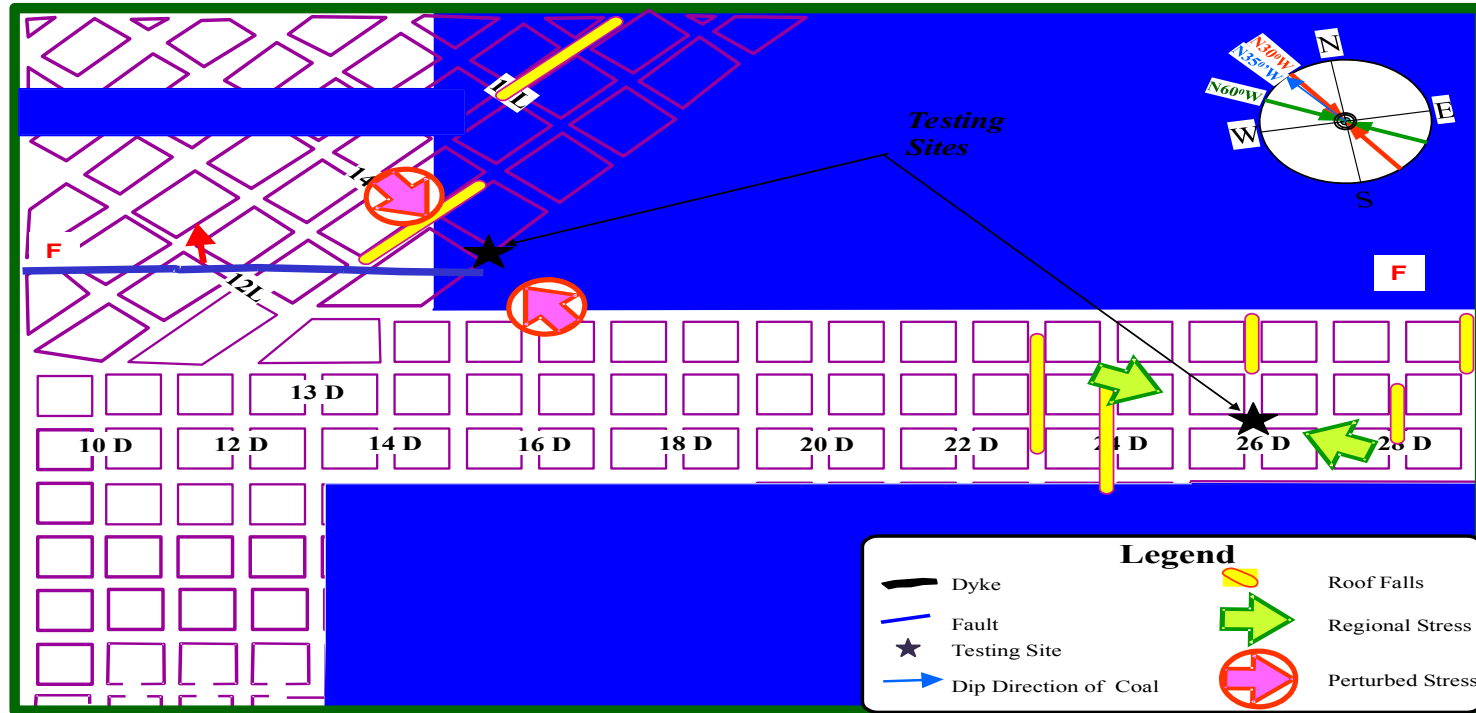
-4.000E+07

-2.000E+07

nirm
kgf



Effect of Faults



Methods of In-situ Stress Measurements

	Method	Volume (m ³)
Hydraulic Methods	Hydraulic fracturing	0.5-50
	Sleeve Fracturing	10 ⁻²
	Hydraulic tests on pre-existing fractures HTPF	1-10
Relief Methods	Surface relief methods	1-2
	Undercoring	10 ⁻³
	Borehole relief method (overcoring BH, slotting etc.)	10 ⁻³ – 10 ⁻²
	Relief of a large rock volumes (bored raise , under excavation technique etc.	10 ² – 10 ³
Jacking Method	Flat jack method	0.5 – 2
	Curved jack method	10 ⁻²
Strain Recovery Method	Anelastic strain Recovery (ASR)	10 ⁻³
	Differential strain curve analysis (DSCA)	10 ⁻⁴
Borehole Breakout Method	Caliper and dipmeter analysis	10 ⁻² – 10 ²
	Borehole televiewer analysis	10 ⁻² – 10 ²
Other methods	Fault Slip Data Analysis	10 ⁸
	Earthquake focal mechanism	10 ⁹
	Indirect Methods (Kaiser Effect etc.)	10 ⁻⁴ – 10 ⁻³
	Inclusion in time dependent rocks	10 ⁻² – 1
	Measurement of residual stresses	10 ⁻⁵ – 10 ⁻³